

Project T27

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All of our tables are 3NF/BCNF Except:
supplied_from We will use Brenstein's Algorithm

① Given supplied_from (supplier_id, drug_name, stock, restock_alert
(I, D, S, R)

$$FD: \begin{cases} ID \rightarrow SR \\ S \rightarrow R \end{cases}$$

② Remove redundant dependencies

keep $ID \rightarrow S \rightarrow ID^+ = \{I, D, R\}$ not redundant

remove $ID \rightarrow R \rightarrow ID^+ = \{I, D, S, R\}$ derived all attribs, redundant

keep $S \rightarrow R \rightarrow S^+ = \{S\}$ not redundant

③ Remove redundant attribs in composite keys

$$\begin{array}{l} ID \rightarrow S \\ S \rightarrow R \end{array} \rightarrow \begin{cases} I^+ = \{I\} \\ D^+ = \{D\} \end{cases} \quad \begin{array}{l} ID \text{ are both necessary to get } S \text{ and } R \end{array}$$

④ Find possible candidate keys

$$\begin{array}{l} ID \rightarrow S \\ S \rightarrow R \end{array} \quad \text{part of key: } \begin{cases} \text{not LHS} \wedge \text{not RHS} = \{\} \\ \text{LHS} \wedge \text{not RHS} = \{I, D\} \end{cases}$$

$$\text{NOT part of key: } \{ \text{LHS} \wedge \text{RHS} = \{S\} \}$$

∴ possible keys are ID, IDR

⑤ Determine which is actual keys

$ID^+ = \{I, D, S, R\}$ key

$IDR^+ = \{I, D, R, S\}$ not key $\rightarrow$ superkey of ID

⑥ Make relations for each FD

$ID \rightarrow S : R_1(I, D, S) \rightarrow R_1(\underline{\text{supplier_id}}, \text{drug_name}, \text{stock})$

FD: $\{\text{supplier_id}, \text{drug_name} \rightarrow \text{stock}\}$

$S \rightarrow R : R_2(S, R) \rightarrow R_2(\underline{\text{stock}}, \text{restock_alert})$

FD: $\{\text{stock} \rightarrow \text{restock_alert}\}$

$R_3(I, D) \rightarrow R_3(\underline{\text{supplier_id}}, \text{drug_name})$

for lossless join $\uparrow$

no FD (similar to medication table)

R_1, R_2 , and R_3 are the final BCNF tables for supplied_from. //